

MINHAZUL ISLAM

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 Personal Website |  GitHub |  LinkedIn

School of Software Technology, Zhejiang University, Ningbo, Zhejiang, China

RESEARCH INTERESTS

I study human AI collaboration in domains where an unsupported claim carries real cost mental-health peer support, health communication, and expert decision-making. My work centers on verifiable language-model output (grounding, citation enforcement, and evaluation methodology), interactive and visual interfaces that make AI systems transparent and controllable, and AI-mediated communication that preserves rather than replaces human judgment and voice.

EDUCATION

- **Zhejiang University** Sept. 2023 – Dec. 2026 (Expected)
M.Eng. in Industrial Design Engineering Ningbo, China
 - **Thesis:** BEPSBot: A Dual-Mode Writing Assistant for Online Bipolar Disorder Peer Support Communities. Advisor: Prof. Mengru Xue.
- **Yunnan University** Sept. 2019 – June 2023
B.Eng. in Computer Science and Technology Kunming, China
 - **GPA:** 3.6/4.0
 - **Thesis:** Product Recommendation System Based on a Collaborative Filtering Algorithm. Evaluated matrix-factorization and similarity-based approaches on goodbooks-10k.

PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Conference Papers

- [C.1] Minhazul Islam, Mengru Xue, and Tasnim Afra (2025). *Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework*. In *Entertainment Computing — ICEC 2025, IFIP TC 14 Workshops*, Tokyo, Japan, pp. 61–75. doi:10.1007/978-3-032-02534-0_8
- [C.2] Tasnim Afra, Mengru Xue, and Minhazul Islam (2025). *Enhancing Biofeedback Interventions for Depression and Anxiety Through Entertainment Computing: A Systematic Review*. In *Entertainment Computing — ICEC 2025, IFIP TC 14 Workshops*, Tokyo, Japan, pp. 37–51. doi:10.1007/978-3-032-02534-0_6

Conference Presentations & Non-Archival Papers

- [P.1] Minhazul Islam, Mengru Xue, and Tasnim Afra (2026). *BEPSBot: Draft-Grounded AI Writing Assistance for Bipolar-Related Online Peer Support*. Oral presentation, HHME 2026 PCC, Zhejiang University, China. [\[conference page\]](#)
- [P.2] Tasnim Afra, Mengru Xue, and Minhazul Islam (2026). *How Should Voice Agents Respond to Anger and Sadness? A Comparison of Empathic Response, Affect-Neutral Acknowledgment, and Cognitive Reappraisal*. Oral presentation, HHME 2026 PCC, Zhejiang University, China. [\[conference page\]](#)

RESEARCH EXPERIENCE

- **Zhejiang University — Graduate Research Assistant** Sept. 2023 – June 2026
Advisor: Prof. Mengru Xue Ningbo, China
 - **Human-AI collaboration for mental-health peer support (BEPSBot).** Designed a draft-grounded generative writing assistant that preserves a peer-support provider’s semantic content while using AI to improve expression, studying how generative assistance can increase utility without displacing human authorship.
 - Conducted a controlled study with 24 peer-support providers; draft-grounded generation increased recommendation adoption from 33.3% to 75.6%, achieved a mean SUS score of 86.98, and preserved improvements in user confidence and satisfaction.
 - Built a dual-mode pipeline combining contextual RoBERTa assessment, sentence-embedding retrieval with ChromaDB, retrieval-grounded LLM generation, strategy-constrained rewriting, and pre/post-generation safety screening.
 - **Clinician-centered AI for bipolar-disorder monitoring.** Conducted semi-structured interviews with seven mental-health professionals and thematic analysis to identify physiological, behavioral, and psychosocial signals relevant to longitudinal bipolar-disorder monitoring; translated findings into design requirements for clinician-centered multimodal AI decision support [C.1].
 - **Research through design.** Designed and prototyped SPROUT, a shape-changing meditative artifact targeting fear of failure in students, combining embedded sensing and actuation with a growth metaphor to explore how tangible interaction can support reflective practice.

APPLIED RESEARCH & ENGINEERING EXPERIENCE

• Synexia AI

AI Infrastructure & Systems Engineer

2025 – Present

Ningbo, China

- Architected two production agentic-AI systems — *Zhanlu*, a governed multi-agent runtime, and *EDIA*, a 17-node LangGraph decision-intelligence system — supporting complex human decision workflows through structured planning, tool use, verification, and evidence-grounded generation.
- Built agent-runtime mechanisms including plan-first execution, sub-agent workflows, goal contracts and bounded self-correction, LLM-as-a-judge evaluation, tool/skill/MCP gateways, and context-isolation controls for reliable multi-agent behavior.
- Reduced EDIA SQL round trips from 77 to 2–5 and retrieved rows from roughly 180,000 to 50–200 through intent-driven query planning and deterministic metric pre-computation, reducing end-to-end latency from 90–478 s to 30–45 s while removing a class of numerical hallucinations.
- Built verifiable-output infrastructure with golden-eval pipelines, citation-grounding checks, hybrid SQL/vector retrieval (ChromaDB, BAAI/bge-m3), and deterministic validation gates that block unsupported claims from reaching users.
- Designed a privacy-preserving dual-LLM architecture in which an on-premise Qwen model reads and sanitizes sensitive enterprise records before external reasoning is allowed, with external routing blocked unless privacy conditions are satisfied.

• Piliving

Technical Product Manager (concurrent, part-time)

June 2024 – Sept. 2024

Ningbo, China

- Analyzed 100,000+ user forum posts and comments using text mining and qualitative categorization to identify recurring product failure modes and translate user evidence into design requirements and prototype revisions.

• ProFabx

Software Engineer (concurrent, part-time)

June 2024 – Sept. 2024

Ningbo, China

- Automated parametric CAD workflows through the Autodesk Inventor API in Python, reducing manual modeling time by roughly 40% and supporting rapid design-space exploration through a browser-based parameter interface.

SELECTED TECHNICAL PROJECTS

• InclusiveVision — Assistive Smart Glasses for Visually Impaired Users

Zhejiang University

Dec. 2023



- Designed and iteratively prototyped an assistive wearable providing real-time auditory obstacle feedback, integrating ultrasonic sensing, embedded interaction, and 3D-printed form-factor iterations for wearability.

HONORS & AWARDS

- **United Nations Technology Bank — Global Youth Talent Program**, selected internationally among young designers 2023

TECHNICAL SKILLS

- **Research Methods:** Controlled user studies, semi-structured interviews, thematic and qualitative coding, research through design, text mining of large social corpora, benchmarking and ablation, usability and user-facing evaluation
- **Human-AI & Interactive Systems:** Human-AI collaboration, agentic decision support, interactive dashboard design, human-centered prototyping, React.js, D3.js, Recharts
- **Multi-Agent & Agentic AI:** LangGraph (17-node FSM), agent harness/runtime design, plan-first execution, sub-agent workflows, goal contracts, bounded self-correction, tracing and observability layers, MCP, tool/skill gateways, LLM-as-a-judge evaluation
- **Context Engineering & Retrieval:** Context isolation and spill management, hybrid SQL/vector RAG, ChromaDB, BAAI/bge-m3, context trimming, deterministic guardrails, provenance and output validation
- **AI Infrastructure & LLM Ops:** FastAPI, SSE streaming, Docker (14-service stack), vLLM, Qwen3-27B, Ollama, PostgreSQL, Redis, MinIO, multi-provider routing, dual-LLM security gates, Prometheus/Grafana, systemd, Nginx
- **Programming & ML:** Python (asyncio), C/C++, SQL, JavaScript, PyTorch, Transformers, scikit-learn, XGBoost, LoRA/QLoRA/PEFT, Git, GitHub Actions CI/CD
- **Languages:** Bengali (native), English (fluent), Mandarin Chinese (conversational)